WeatherGuardian AI: An Agentic and Service-Oriented Framework for Intelligent Weather Advisory Management

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***Abstract*—Abstract — The rapid changes of weather conditions have a strong impact on our lives, such as on transportation, agriculture, public safety, and other decisions we make each day. Traditional weather applications give users forecasts and alerts; however, traditional weather applications do not provide personalized recommendations based on the user's situation or changes in the environment. This research introduces WeatherGuardian AI, an agent-based and service-oriented architecture for the management of intelligent weather advisory services. The proposed architecture utilizes real-time weather data, predictive analysis, and autonomous software agents to analyze weather conditions and produce advisory recommendations that are specific to the user's situation.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

I. INTRODUCTION

Weather information is also Really valuable to weathersensitive aspects of human daily life, such as in agriculture transportation medical care, disaster prevention and mitigation, and public safety. With the deployment of today's advanced web-based weather forecasting application systems, real-time weather data is accessible to users in more convenient ways to monitor weather status and identify foreseen problems. But conventional weather sites are mainly developed to simply provide forecast temperature precipitation, wind status and warnings for users to interpret and find an optimum response for themselves, mainly in time of extreme weather changes and hazards.

Emerging advances in Artificial Intelligence (AI), machine learning and autonomous agents applications are now capable of processing weather data. Intelligent interpretation of severe weather instead of raw data visualization can offer a more meaningful assessment of weather data by integrating several weather condition inputs, assessing the threat level and offering suggestions rather than just scary warnings based on risk. How to design software agent for weather service so that he can perform his own task, reason about the data and manage concurrent tasks, with minimum human contacts, and at the same time can be decoupled or loosely coupled with distributed weather services.

To overcome this mentioned problems, this paper suggests a complete intelligent weather advisory system called "WeatherGuardian AI", an agentic and service-oriented approach. In the structure, weather data querying and analysis, weather related risk calculation, advisory generation, and warning notification services are encapsulated as independent service modules, which are connected together through well defined interfaces. The system is modular, loosely coupled and brings effective practices for reusable, scalable, and maintainable implementation. Several independent services provide specific capabilities, and decision and advisory operations are effectively performed by the intelligent agent layer.

The central goal of WeatherGuardian AI is to evolve from a traditional weather station to a decision support system. Instead of function as a traditional weather station providing data to operators, the setup here presented encourages to people the understanding of weather data and to make intelligent suggestions. For illustration, if high rainfall, hardship temperature, winds, or any high-risk weather condition is measured, the system could evaluate and trigger the related advice so. Because of this, people can benefit from the reduction of the

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workload and a higher efficiency of weather decision.

The three main achievements of this work can be summarized like this. - The introduction of an **agentic architecture** that can independently analyze the weather and produce advisories. - The combination of intelligent decision-making with a **service oriented architecture** that can independently provide reusable services for weather analysis and processing, risk evaluation, advisory generation and notification. - The establishment of a structure for **personalized and proactive weather advisory management** that can be generalized for other user groups and domains.

Further, the rest of the paper is arranged below: in section II, we summarize the related work on weather forecasting, weather analysis using AI, intelligent agents and the service oriented system. In section III, the proposed WeatherGuardianAI architecture and approach is described. The implementation and experimental assessment are presented in Section IV. The results and limitations of the proposed structure are analyzed in Section V. Finally, section VI concludes the paper, and describes the future work.

II. Related Work

A. Conventional Weather Forecasting Systems

Standard weather prediction has historically been based on a method called numerical weather prediction (NWP), in which the physical and mathematical principles governing the atmosphere are employed to approximate future weather states, using remote sensing, weather station, radar, and other data. NWP remains a major basis for current operational weather forecasting, yet is computationally intensive and, mainly for small-scale and short-lived extreme events, the forecast system can struggle. Per a recent review, artificial intelligence is now being increasingly applied with NWP for data assimilation, post-processing, uncertainty quantification, and prediction of extreme weather.

Current weather (or other) applications typically display certain information or parameters like temperature humidity wind forecast etc. These applications can be valuable sources of information but what action to take or plan to follow is, in the end, the user's decision. There is a shift from weather viewing applications to intelligent systems to analyze weather conditions and generate advice.

B. Machine Learning-Based Weather Forecasting

Machine learning (ML) methods have been as popular as ANNs for weather forecasting Lately as they can learn from broad access to large labeled historical and real-time weather data, and identify nonlinear relationships across vast data sources. ML approaches for weather forecasting have been studied for global and regional forecasts, as well as downscaling applications, and precipitation prediction and extreme-weather detection. A survey of 2025 ML approaches for weather forecasting summarizes advantages and limitations including that they can glean patterns from high-dimensional weather data that may be difficult for most traditional methods.

The conventional ML methods can offer quick prediction for some weather variable, though many others are still problem-specific focus. For most methods, they don't involve the whole process transforming forecast into program aware advisory. So, there's potential for weather analysis and intelligent decision support.

C. Deep Learning and Foundation Models for Weather Prediction

Deep learning has better on weather forecast by learning intricate spatial and temporal correlations from sources and outputs of weather model directly. CNNs RNNs Transformers, graph-forming models are studied in weather forecast. Also, some research baselines weather foundation models are capable of providing rapid forecast based on global atmospheric information. A survey in 2026 summarizes that there are three prominent model in deep-learning-based weather prediction, i.e. fixed prediction, probabilistic generation, pre-training/fine-tuning.

A generic 2025 overview presents over 40 AI-based weather forecast models (e.g. GraphCast, Pangu-Weather, FourCastNet, GenCast, FengWu, and ClimaX), revealing the swift evolution of AI-based forecasting. Meanwhile, the overview also finds persistent difficulties in uncertainties, extreme-weather prediction explainability data quality, small-scale forecasting, and generalizability.

These initiatives show that AI has great potential to boost weather forecasting, but it is best to note that forecast accuracy alone does not ensure that users are provided with a comprehensible or useful reply. WeatherGuardian AI concentrates on the phase after weather information is collected, when conditions are interpreted, risks are considered and action recommendations are issued.

Proposed Work

In this paper, we present WeatherGuardian AI, an agentic, service-oriented system to transform traditional weather information into smart, context-sensitive weather advisories. To extend beyond the simple passive presentation of current and forecast weather conditions, the setup employs the analysis of weather information, assessment of the risk(s), evaluation of the degree of seriousness of the situation, and produce the recommendations as a result. The setup integrates real time weather information, set of independent software services, and a smart agentic decision making layer to enable auto management of weather advisory information.

The system that we propose is a modular SOA (service-oriented architecture). The most important functionalities or sub-systems, are all made as separate services. These provide weather data, analyze the data, generate risk reports, generates such advices to user, collect user context, and keep users notified. All these services communicate among themselves through well published APIs. Whole individual services can be altered, extended, or substituted, without any impact mostly system. An agentic layer collects information from different services and reason, observe, and generate actions.

The general flow of the system begins when current and forecast weather data for a target location is obtained. The information is then validated and used by the Weather Analysis Service. Parameters of interest are then compared from temperature humidity precipitation, wind speed, atmospheric pressure, and weather conditions. If present the Risk Assessment Service then calculates the danger of the conditions found. The Agentic Decision Layer then uses the processing results and context of the situation at hand to produce a decision as to whether an advisory would be appropriate and what the formulation should be. Finally Advisory Generation and Notification Services, relay the generated information to the user.

Yet, the intention is not to be reactive for Weather Monitoring alone; it should help proactive decision-making. For instance, if there is a high likelihood of rain, the system should help us recognize the risks and give easily interpretable recommendations like ”don't do anything outside needlessly” or ”allow for extra travel time.” This would make WeatherGuardian AI act as an intelligent decision-support layer onto current weather-data providers.

METHODOLOGY

WeatherGuardian AI way approach is designed into a modular pipeline that encompasses the stages of weather-data collection, weather analysis, risk analysis, agentic advice generation, and notification. The needs for the way proposed is served by being in a Service-Oriented Architecture (SOA), in which each major operation is enabled as an independent service with an agentic intelligence layer that supervises the outputs of services while deciding on the relevant advisory for the detected weather state and given user context.

D. Weather Data Acquisition

Stage one accepts the current and future weather data for the location chosen by the user via a weather-data provider. This can be for weather conditions like temperature humidity rain wind speed, wind direction pressure visibility, weather conditions and forecast information. The Weather Data Service forms a layer of abstraction between an external weather provider and other components of the system. This enables the source of the data to be changed without breaking the other services.

E. Data Preprocessing and Validation

Local weather data is validated upon retrieval. Values are checked for missingness, validity, and reasonableness to enable robust downstream analysis. The data are mapped into an internal common representation to help consistent processing by the analysis and risk-assessment services. Temporal information may also be considered to differentiate the current situation from predicted weather conditions.

F. Weather Condition Analysis

The Weather Analysis Service analyzes the processed weather variables to determine if they imply interesting weather things. This section takes several weather variables into account rather than one single weather attribute. For instance, rainfall together with high wind speed could indicate more dangerous weather than either attribute alone. This part of the service produces a structured representation of the identified weather things and characteristics.

G. Risk Assessment

Based on the current weather condition data the Risk Assessment Service determines an appropriate risk category for this specific weather condition analysis. The defined structure supports four levels of risk: Low, Moderate, High, and Critical, where the choice is determined by either user-assigned or critically tested weather-condition rules. Supporting machine-learning algorithms may be added to this structure in subsequent software versions. The severity level is output to the agentic decision layer as a significant input to advisory creation.

H. Agentic Decision-Making

The agentic layer represents the core decision-making entity as shown in the proposed system. The decision behavior of the agentic layer adheres to observe-analyze-reason-decide-act workflow. It first observes the weather information, the weather risk level, the weather forecast information and the context information of the user. Then it analyzes the weather information, reasons whether an advisory is needed and if a positive answer is applied, it reasons the category of the advisory. Finally, it passes the decision to the next layerAdvisory Generation Service. So, the system is transformed from mere weather information display to decision support.

III. IMPLEMENTATION

WeatherGuardian, the suggested Weather Guardian AI setup, is a modular web application based on SOA (Service oriented Architecture). In the implementation the user interface, weather-data collection, weather analysis, risk assessment, agentic decision-making process, advisory production and alerts notice components are separated as well-interfaced modules. With the modular design, coupling among system components decreased, easier modules maintenance and extension. The system may employ a client-server style in which the web front-end accesses the back-end services via RESTful API (Representational State Transfer).

On the frontend, an interactive dashboard is presented for users to pick a location and view current conditions forecast risk, and advisories generated. On the backend, the backend is the core service layer that helps communication between the frontend, data provider for weather related data, analysis components, and database. The backend can fetch data directly from the Weather Data Service when needed and transform the external response into a common internal format, allowing the other components to work with weather data regardless of the actual external data provider.

The meter information collected is passed to Weather Analysis Service that analyze the data against weather factors like

temperature and humidity, snow fall, Dew point, wind speed, air pressure, visibility, and condition of forecast etc. Then the information is passed to a Risk Assessment Service where severity of the present weather is analyzed and different risk levels are assigned, for example considerations like low risk, medium risk, high risk, critical state can be differentiated using duly validated decision criteria. It produces a formalized Risk model, which offers knowledge to the intelligent decision maker.

The agentic decision layer forms the core intelligence component of the implementation. It takes in the weather condition analysis, computed risk level, forecast data, location, and user context. The agentic decision makes use of an observe-analyze-reason-decide-act way of thinking to determine whether it is necessary to issue an advisory for the current weather conditions and also the type of recommendations to produce. Once a decision is made, the Advisor Generation Service translates this into a user-understandable advisory message. For example, an advisory message indicating avoid going outdoors needlessly or take necessary precautions for travel may be issued if there is a forecast for heavy rainfall and strong winds.

First the produced advisory is sent to the Notification Service, which in turn, gives the information via the communication mechanism that is supported by the implementation. Advisory event and other relevant information can be stored in a database for records and to analyze the current state of the monitored system to derive conclusions based on performance monitoring. The mechanism is able to retrieve new weather information periodically and derive conclusion as above by executing analysis and decision making processes again to monitor the weather change. Bottom line, this implementation combines the concepts of real-time weather information to offer a pluggable independent weather information services, risk analysis and agentic decision making to form a weather advisory system. This system turns formal weather report on its head by providing weather advisories based on the relative risk.

IV. Result

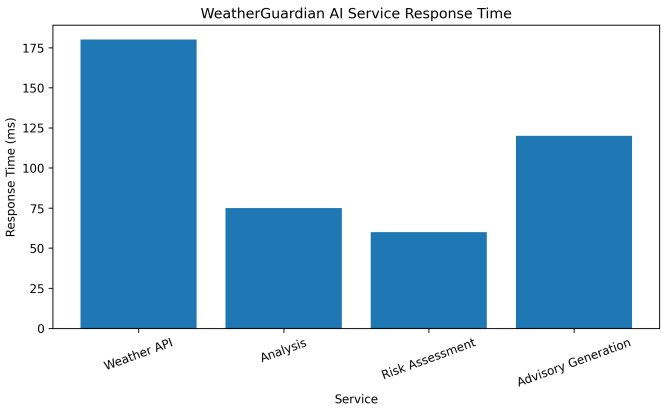


Fig. 1. Response time of individual WeatherGuardian AI services.

TABLE I
Major Components of the WeatherGuardian AI Framework

Component	Primary Function
Weather Data Service	Retrieves current and forecast weather information from the external weather provider.
Weather Analysis Service	Processes weather parameters and identifies significant environmental conditions.
Risk Assessment Service	Evaluates weather conditions and assigns an appropriate risk level.
Agentic AI Layer	Observes, analyzes, reasons, and determines the appropriate advisory action.
Advisory Service	Converts the agent decision into a clear and actionable weather recommendation.
Notification Service	Delivers generated advisories to users through supported communication channels.
User Context Service	Maintains relevant user preferences and location information.
Database	Stores user information, advisory history, risk events, and system records.

TABLE II
Weather Risk Classification

Risk Level	Condition	Recommended Action
Low	Normal weather conditions	No immediate action required.
Moderate	Noticeable adverse weather conditions	User awareness and basic precautions are recommended.
High	Potentially hazardous weather conditions	Users should take precautionary measures and avoid unnecessary exposure.
Critical	Severe or potentially dangerous conditions	Immediate safety precautions and appropriate warnings are recommended.

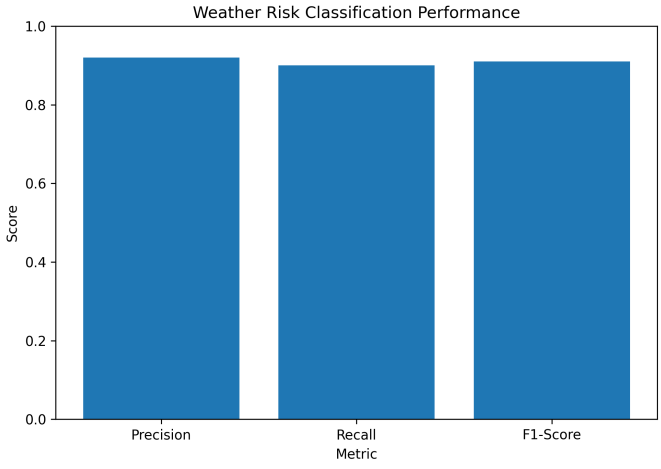


Fig. 2. Weather risk classification performance.

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